Global CO_2 Emissions Analysis

AMS 597 Statistical Computing Project

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1 Introduction

Climate change remains one of the most pressing global challenges, with carbon dioxide (CO_2 emissions being a primary driver. Understanding the relationship between economic development, energy consumption, and emissions is crucial for designing effective environmental policies.
This project investigates global CO_2 emissions using a multi-country, multi-year dataset. The analysis focuses on identifying key economic and energy-related factors influencing emissions, uncovering structural patterns across countries, and evaluating predictive and statistical models.
The study is structured around four research questions (R1–R4), covering regression analysis, clustering, forecasting, and energy efficiency dynamics. While the later sections address modeling and inference, this report begins with data understanding through preprocessing and exploratory data analysis (EDA), which form the foundation for all subsequent analysis.

2 Dataset Description

The dataset used in this project is obtained from Our World in Data and contains global information on CO_2 and greenhouse gas emissions across countries and years 1750 to 2024.
It includes over 40,000 observations and more than 70 variables, covering environmental, economic, and energy-related indicators such as: CO_2 emissions and emissions per capita, GDP and population, Energy consumption metrics, Methane and other greenhouse gases.
After preprocessing, a cleaned subset of approximately 2,640 country-year observations with 20 variables was used for analysis. The dataset contains both:

- **Continuous variables**: CO_2, GDP, energy consumption, population
- **Categorical variables**: country

The dataset is particularly suitable for this project due to:

- Its large size and global coverage
- Presence of missing values requiring preprocessing
- Rich combination of economic and environmental indicators

This allows for meaningful statistical analysis of how economic activity and energy usage influence emissions across countries and over time.

3 Data Cleaning and Preprocessing

The raw dataset required substantial preprocessing before analysis. The key steps are:

3.1 Handling Missing Values

The dataset contained missing values across several variables due to inconsistent reporting across countries and years. Observations with excessive missingness were removed, and only variables with sufficient data availability were retained.

3.2 Filtering and Subsetting

- Only observations from year 1990 onward were included
- Invalid or missing country identifiers were removed
- A subset of 20 relevant variables was selected for analysis

3.3 Feature Selection

Variables directly relevant to emissions, energy usage, and economic activity were retained:

- GDP, population
- Energy consumption metrics
- CO_2 emissions and related components (coal, oil, gas)
- Other greenhouse gases (methane, nitrous oxide)

3.4 Data Consistency

Care was taken to ensure:

- Consistent units across variables
- Removal of duplicate entries
- Alignment of country-year observations

The resulting dataset is clean, consistent, and suitable for statistical modeling and EDA.

4 Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to understand the structure of the dataset, identify patterns, and guide modeling decisions.

4.1 Distribution of CO_2 Emissions

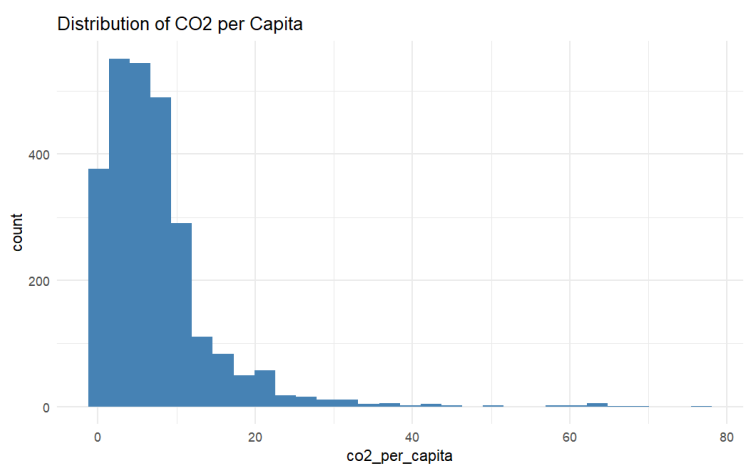


Figure 1: Distribution of CO_2 emissions per capita

The distribution of CO_2 emissions per capita is highly right-skewed. Most country-year observations fall within a low-to-moderate emission range, while a small number of countries exhibit extremely high emissions.

This indicates:

- Presence of significant outliers
- Strong inequality in emissions across countries

4.2 GDP vs CO_2 Emissions

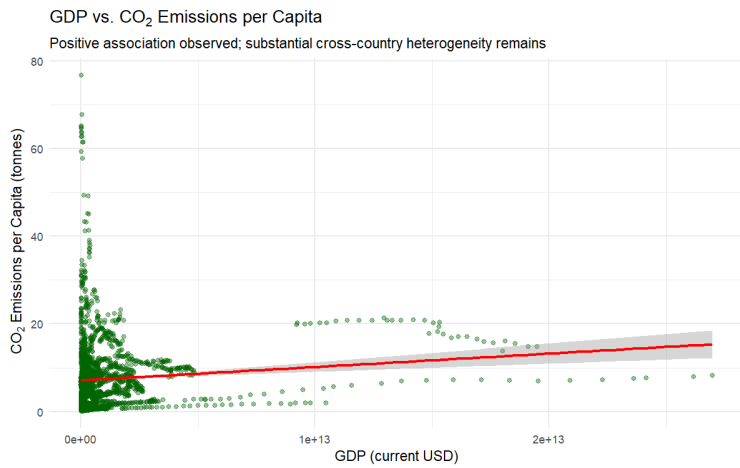


Figure 2: GDP vs CO_2 emissions per capita

A weak positive relationship is observed between GDP and CO_2 emissions per capita. However, the large dispersion indicates substantial heterogeneity across countries. Key insights:

- Higher GDP is generally associated with higher emissions
- The relationship is noisy and not strongly linear
- Economic size alone does not fully explain emissions

4.3 Energy Consumption vs CO_2

A strong linear relationship is observed between energy consumption per capita and CO_2 emissions. This suggests that energy usage is a primary driver of emissions.
Compared to GDP, energy consumption provides a much clearer and stronger predictive signal.

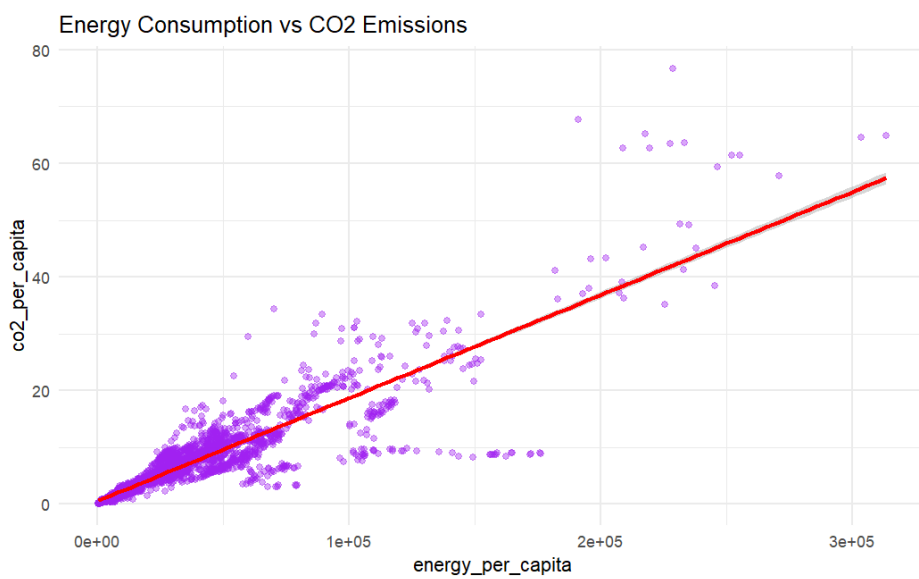


Figure 3: Energy consumption per capita vs CO_2 emissions per capita

4.4 Correlation Analysis

The correlation matrix highlights several important relationships:

- Strong positive correlation between CO_2 and energy-related variables

- High correlation among different emission sources (coal, oil, gas)
- Moderate correlation between GDP and emissions

This suggests potential multicollinearity among predictors, which motivates the use of regularization methods in later modeling.

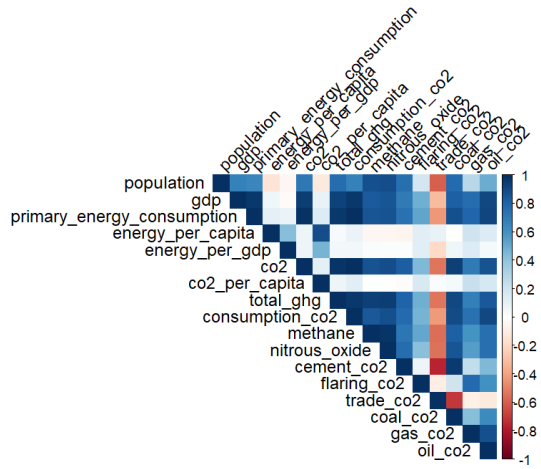


Figure 4: Correlation matrix of key variables

4.5 Global Emissions Trend

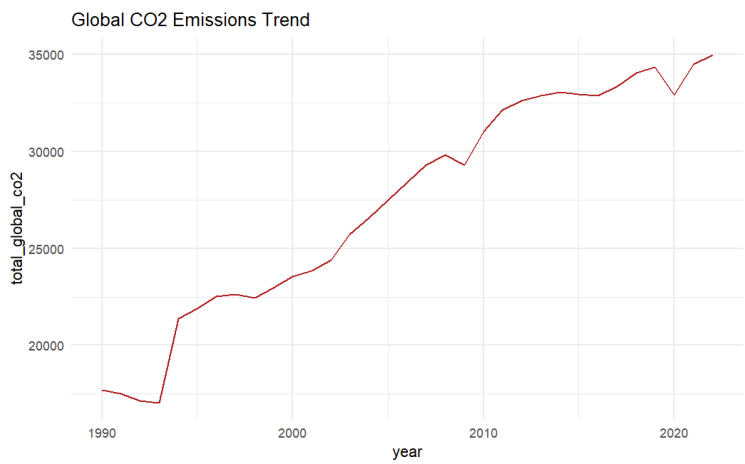


Figure 5: Global CO_2 emissions over time

Global CO_2 emissions show a strong upward trend over time, with noticeable dips around major global events such as economic recessions and the COVID-19 pandemic.

This indicates:

- Long-term growth in emissions
- Sensitivity of emissions to global economic activity

Summary of EDA

The exploratory analysis reveals several important findings:

- CO_2 emissions are highly skewed with significant outliers
- Energy consumption is the strongest predictor of emissions
- GDP alone provides limited explanatory power
- Global emissions have increased steadily over time

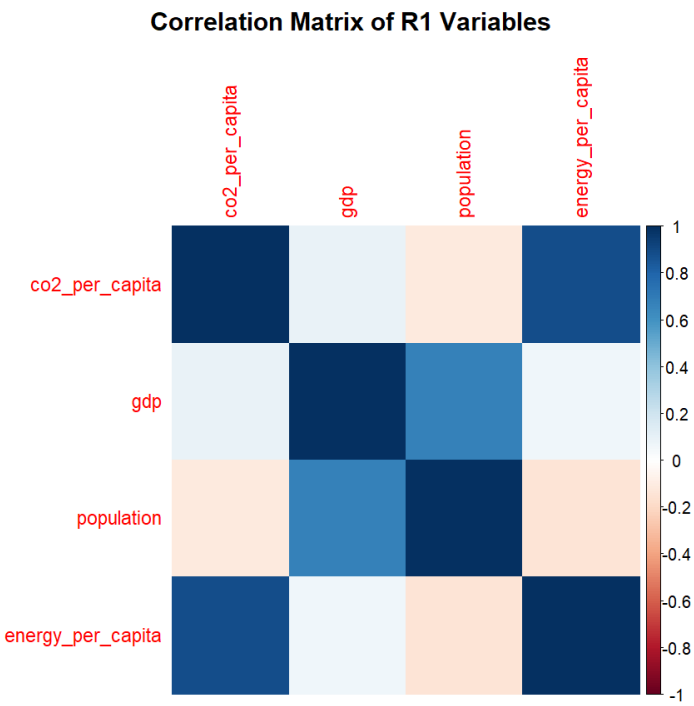
These insights guide the modeling approaches used in subsequent sections of the project.

5 R1: Regression Analysis

5.1 Objective

We have R1: Which economic and energy-related variables are most strongly associated with CO_2 emissions per capita?
The response variable is CO_2 emissions per capita, and key predictors include GDP, population, and energy consumption per capita.

5.2 Correlation Analysis



The correlation matrix reveals that **energy consumption per capita has the strongest positive correlation with CO_2 emissions**. GDP shows a moderate positive association, while population exhibits weak or negligible correlation after accounting for other variables. This suggests that emissions are driven more by *intensity of energy use* rather than sheer population size.

Figure 6: Correlation matrix of variables used in regression

5.3 Model Comparison

Three models were fitted:

- Ordinary Least Squares (OLS)
- Ridge Regression
- Lasso Regression

All three models perform similarly, achieving high R^2 values (above 0.8), indicating strong explanatory power. OLS slightly outperforms regularized models in RMSE and MAE, suggesting that multicollinearity is not severe enough to degrade OLS performance significantly.
Regularization (Ridge and Lasso) provides only marginal differences, implying that the underlying relationships are relatively stable and well-behaved.

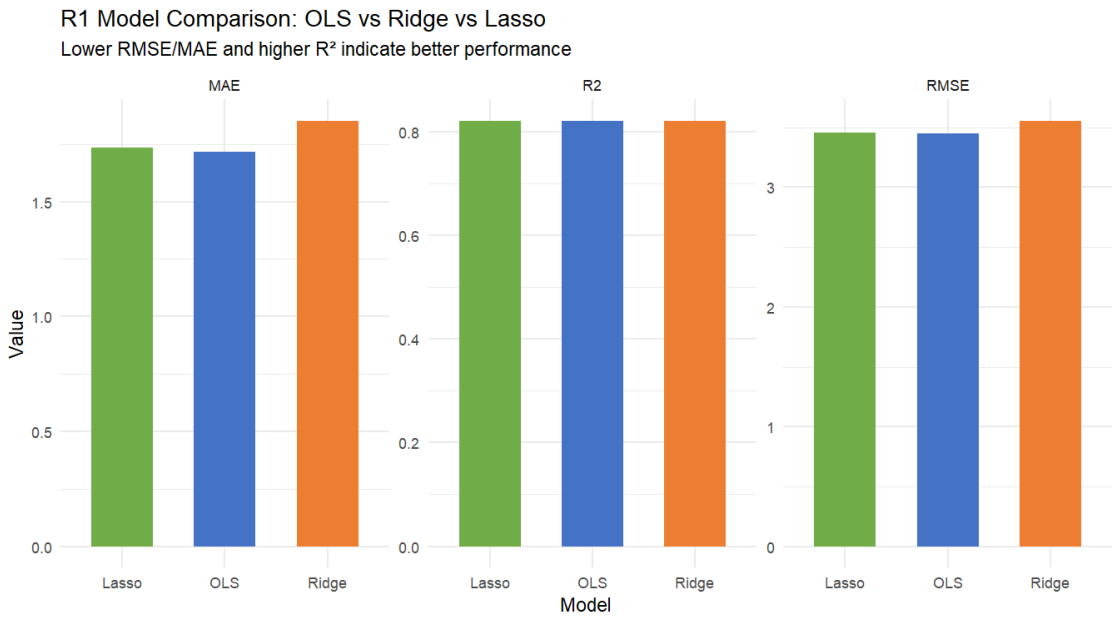


Figure 7: Model comparison using MAE, RMSE, and R^2

5.4 Actual vs Predicted Performance

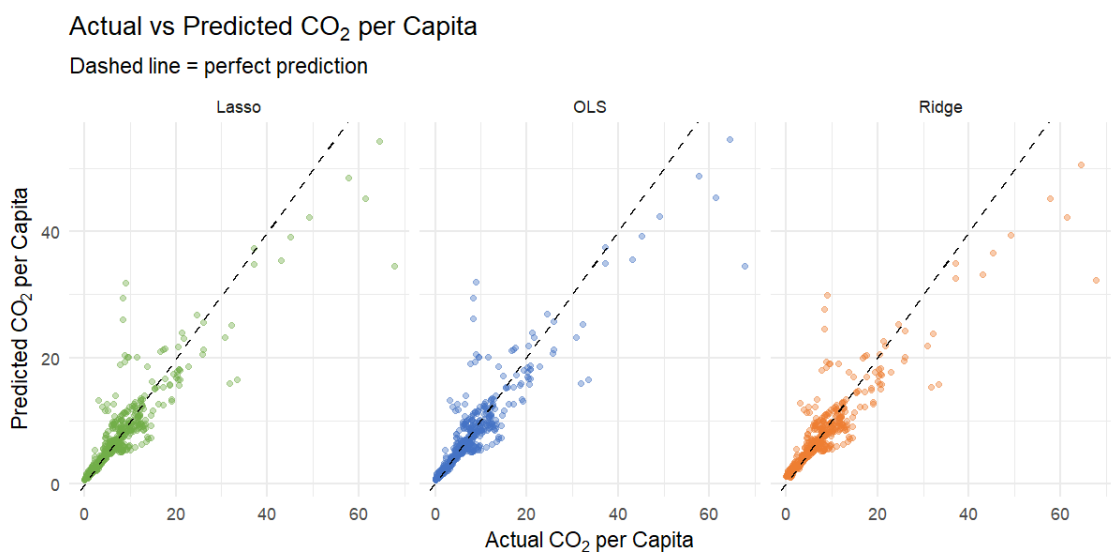


Figure 8: Actual vs predicted CO_2 emissions per capita

Predictions from all models closely align with the 45-degree reference line, particularly for lower and moderate emission values, indicating good predictive performance. OLS and Lasso show slightly tighter clustering around the line, suggesting more accurate predictions, while Ridge shows slightly greater dispersion. However, deviations increase for high-emission observations, indicating reduced predictive accuracy in extreme cases.
This reflects the influence of outliers and structural differences among high-emission countries.

5.5 Model Diagnostics

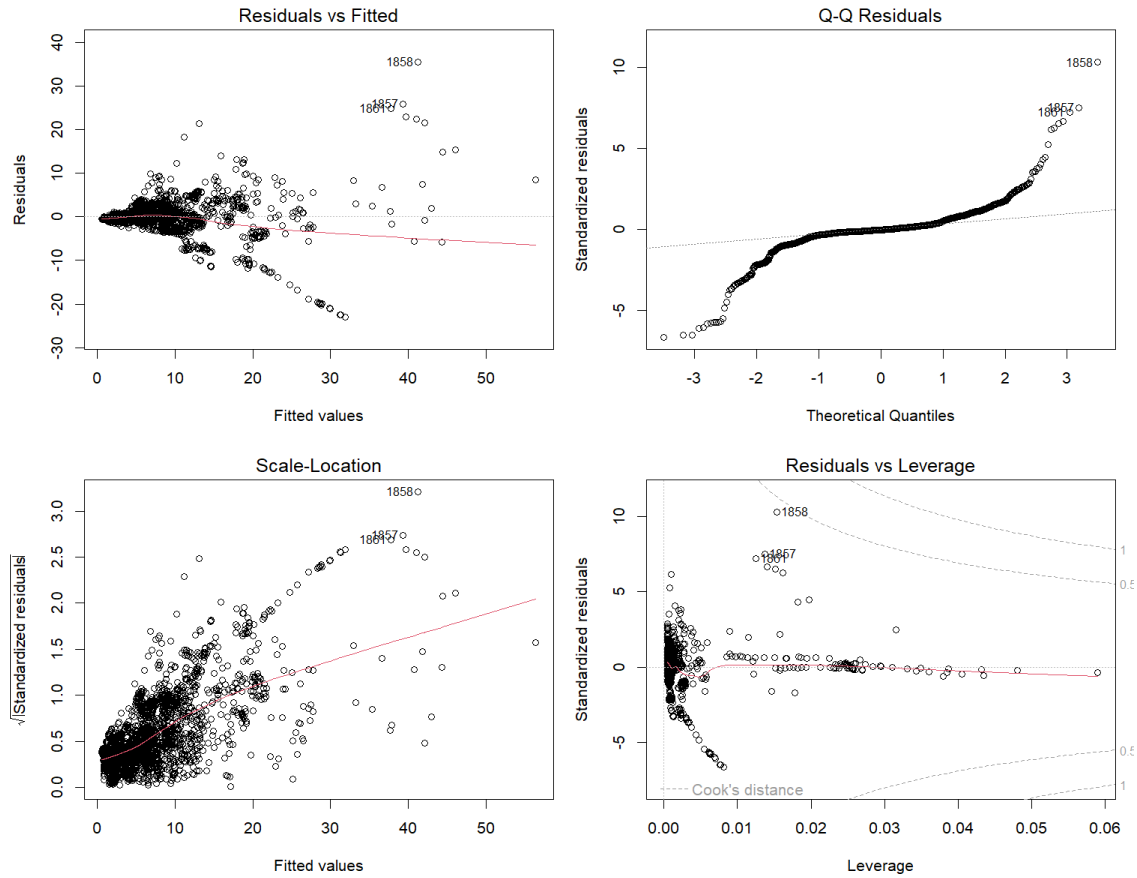


Figure 9: OLS diagnostic plots

Diagnostic analysis highlights several issues:

- **Heteroscedasticity**: Residual variance increases with fitted values
- **Non-normality**: Q-Q plot shows heavy tails due to extreme emitters
- **Influential observations**: A small number of high-leverage points affect model fit

Despite these violations, the model remains useful due to strong overall predictive performance.

5.6 Key Findings (R1)

- Energy consumption per capita is the **dominant predictor** of CO_2 emissions
- GDP has a secondary positive effect
- Population is not a significant predictor after controlling for energy usage
- Linear models provide strong performance, with limited gains from regularization

6 R2: Clustering Analysis

6.1 Objective

We have R2: Can country-year observations be grouped into meaningful clusters based on emission-related characteristics, and do different unsupervised learning methods yield consistent groupings? Clustering was performed using:

- K-Means
- Hierarchical Clustering
- DBSCAN
- Gaussian Mixture Models (GMM)

6.2 Dimensionality Reduction (PCA)

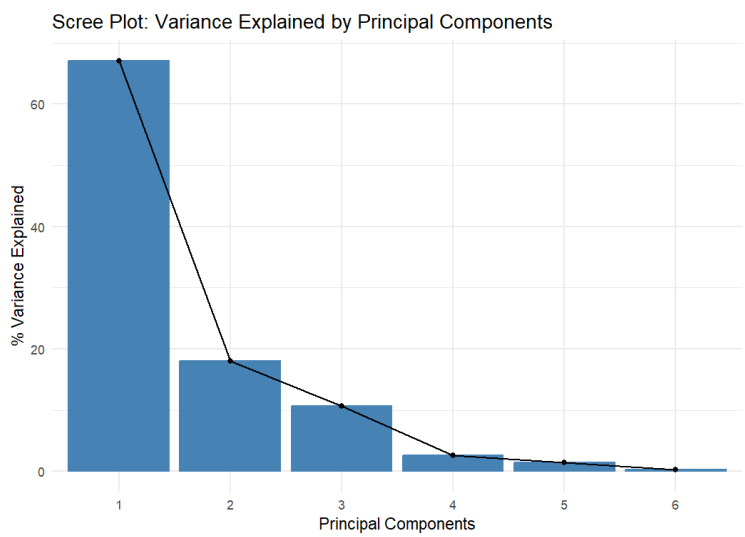


Figure 10: Scree plot for principal components

The scree plot shows the proportion of variance explained by each principal component. The first principal component explains the majority of the variance (around 65–70%), while the second component adds a smaller but still meaningful contribution. Together, the first two components capture a large portion of the total variance, indicating that most of the information in the dataset can be represented in reduced dimensions. The remaining components contribute relatively little, suggesting diminishing returns beyond the first few components.

6.3 Optimal Number of Clusters

The elbow plot shows how the within-cluster sum of squares (WSS) decreases as the number of clusters increases. A sharp decline is observed from k = 1 to k = 3, indicating that adding clusters significantly improves model fit in this range.
Although the silhouette analysis indicates that k = 2 provides the highest clustering quality, k = 3 was selected to balance statistical performance with domain interpretability. Using three clusters enables classification into low, medium, and high emission groups, which better reflects real-world differences in emission levels. While this choice slightly reduces clustering quality, it provides more meaningful and actionable insights.
Using three clusters allows the data to be categorized into low, medium, and high emission groups, which better reflects real-world differences in countries' emission levels. While this choice slightly reduces clustering quality, it provides more practical insights and richer interpretation.

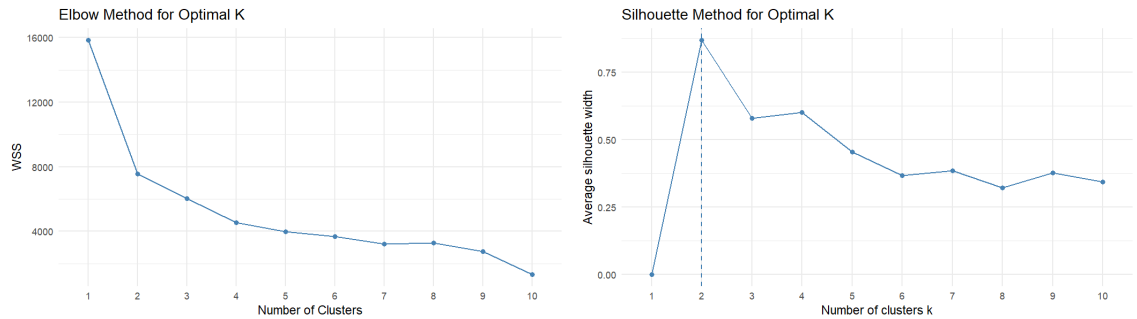


Figure 11: Elbow and silhouette methods for selecting number of clusters

6.4 K-Means Clustering

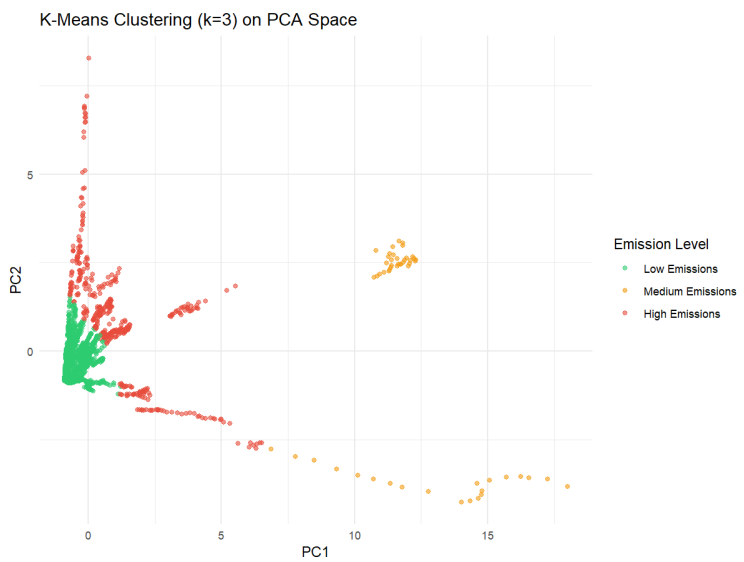


Figure 12: K-Means clustering in PCA space (k=3)

The K-means clustering results (k = 3) visualized in PCA space show clear grouping of observations into three distinct emission categories. The "Low Emissions" cluster forms a dense group near the origin, indicating countries with consistently low emission levels. The "Medium Emissions" cluster appears more separated along the principal components, suggesting moderate but distinct emission patterns. The "High Emissions" cluster is more dispersed, reflecting greater variability among high-emitting countries.

The average silhouette score (0.586) indicates moderate clustering quality, meaning that while clusters are reasonably separated, some overlap still exists between groups. Overall, the visualization confirms that the clustering captures meaningful structure in the data.

6.5 Comparison with Other Methods

The PCA comparison plot illustrates how different clustering methods partition the data into groups. K-means and hierarchical clustering produce very similar cluster structures, with clear separation between low, medium, and high emission groups. This consistency aligns with their high agreement (ARI), indicating stable and reliable clustering.

DBSCAN identifies more clusters with irregular shapes and highlights noise points, demonstrating its ability to capture complex patterns and outliers that other methods do not detect. In contrast,

GMM shows more overlapping clusters, reflecting its probabilistic nature and weaker separation. Overall, the visualization confirms that while K-means and hierarchical clustering provide consistent and interpretable groupings, DBSCAN captures more complex structures, and GMM results in less distinct clusters.

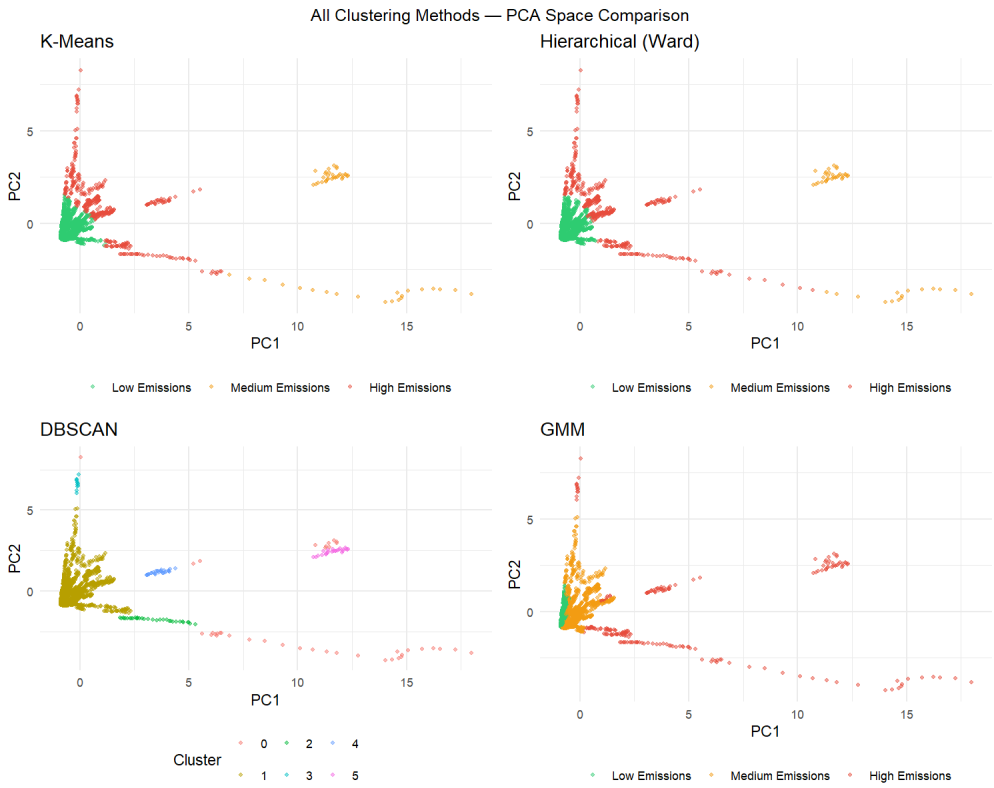


Figure 13: Comparison of clustering methods in PCA space

6.6 Silhouette Analysis

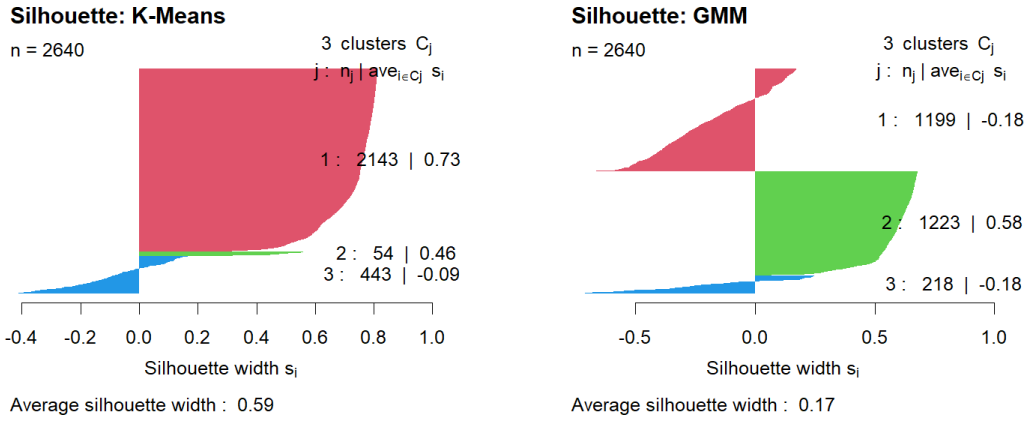


Figure 14: Silhouette plots for K-Means and GMM

The silhouette plots compare the clustering quality of K-means and GMM using three clusters. K-means shows a higher average silhouette score (0.59), indicating better-defined and more cohesive clusters. Most observations have positive silhouette values, especially in the largest cluster, suggesting strong separation.
In contrast, GMM has a much lower average silhouette score (0.17), with many observations having negative values. This indicates significant overlap between clusters and weaker separation.
Overall, the comparison shows that K-means produces more distinct and well-separated clusters, while GMM results in less clearly defined groupings.

6.7 Key Findings (R2)

- Three distinct emission clusters (low, medium, high) are consistently identified
- K-Means provides the most stable and interpretable clustering
- GMM captures probabilistic structure but introduces overlap
- DBSCAN is useful for detecting outliers rather than forming global clusters

7 R3: Forecasting CO_2 Emissions

R3: Can we accurately forecast country-level CO_2 emissions per capita using historical socioeconomic and energy-use features, and which model family generalises best to an unseen future period?

7.1 Objective

The objective of R3 is to evaluate whether future CO_2 emissions per capita can be accurately forecast using a range of statistical, machine learning, and deep learning models, and to assess whether more complex models outperform classical linear approaches.
The target variable is `co2_per_capita`, and a time-based split is used:

- Train: ≤ 2015
- Validation: 2016–2019
- Test: 2020–2022

7.2 Feature Engineering and Leakage Prevention

To ensure a valid forecasting setup, only lagged and safe covariates were used. The following precautions were applied:

- Removed direct CO_2 components (e.g., total_ghg, coal_co2) to prevent leakage
- Created lag features (e.g., CO_2 lag-1, lag-3)
- Rolling averages computed with shift(1) to avoid future information
- Scaling performed using train-only data for neural networks
- TimeSeriesSplit used for stacking instead of random cross-validation

7.3 Models Considered

The following models were evaluated:

- Linear Models: Linear Regression, Ridge, Lasso, ElasticNet
- Tree-based Models: Random Forest, XGBoost
- Neural Networks: MLP, LSTM
- Pretrained Models: PatchTST, TimesFM
- Ensembles: Averaging, Weighted, Stacking

7.4 Model Comparison

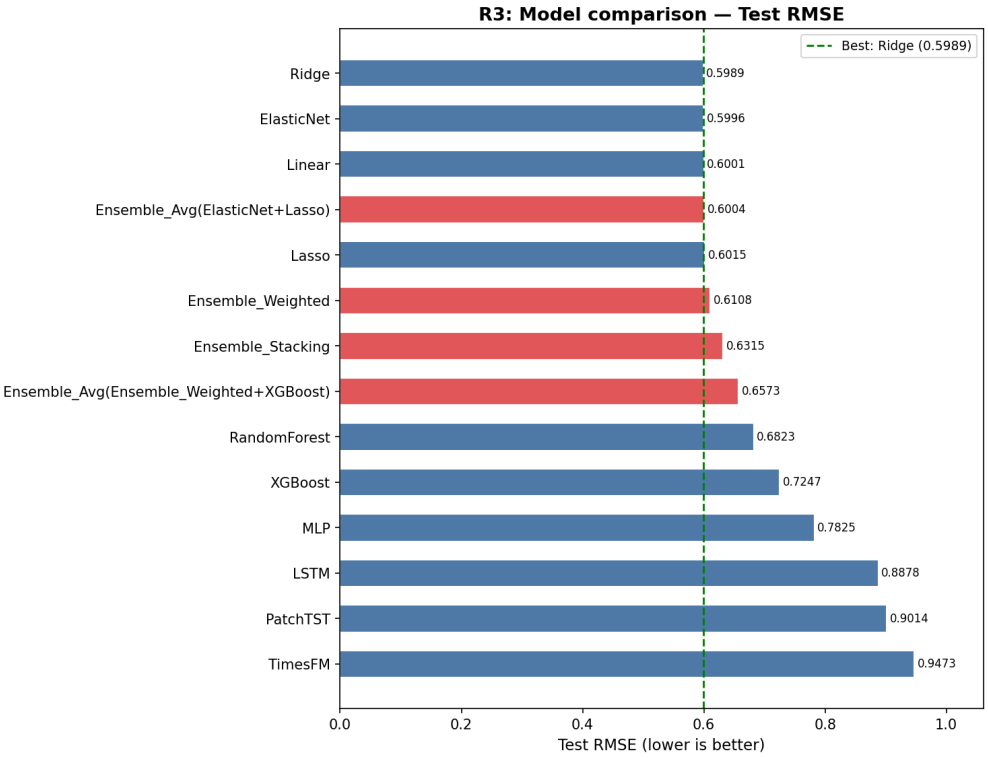


Figure 15: Test RMSE comparison across models (lower is better)

The results show that all four linear models perform nearly identically, with Ridge achieving the best performance.

Model	RMSE	MAE	R^2
Ridge	0.5989	0.3126	0.9891
ElasticNet	0.5996	0.3172	0.9891
Linear	0.6001	0.3072	0.9891
Lasso	0.6015	0.3170	0.9890

Table 1: Top-performing models on the test set

7.5 Best Model: Ridge Regression

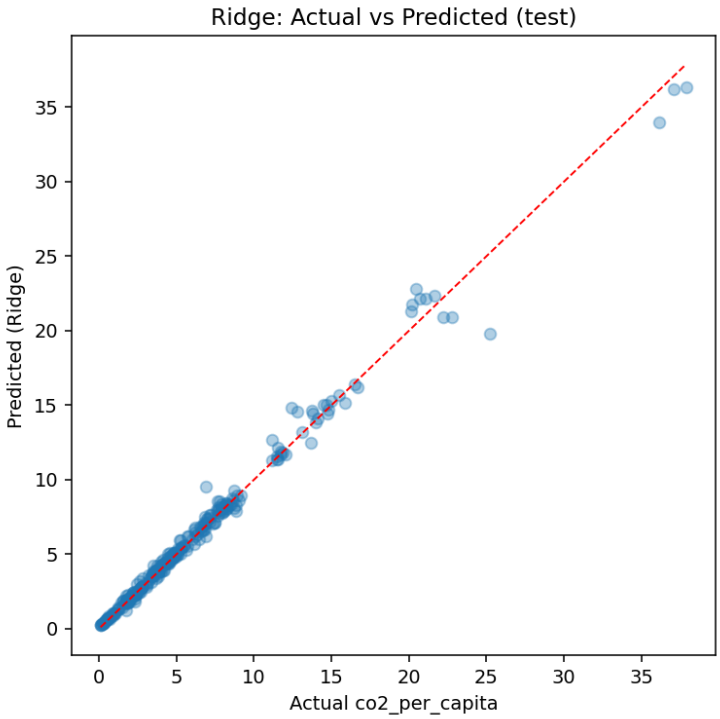


Figure 16: Ridge: Actual vs Predicted CO_2 per capita

Ridge regression achieves the lowest test RMSE (0.5989) and produces predictions closely aligned with the 45-degree reference line, indicating strong generalization and minimal bias across observed values. All four linear models (OLS, Ridge, Lasso, ElasticNet) produced very similar test RMSE values in the range 0.599 – 0.602. Ridge with alpha=1.0 was marginally the best single linear model. This cluster of near-identical performance reflects the fact that with 11 features on a large panel, OLS is already nearly optimal and additional regularisation provides only marginal benefit.

7.6 Comparison with Nonlinear Models

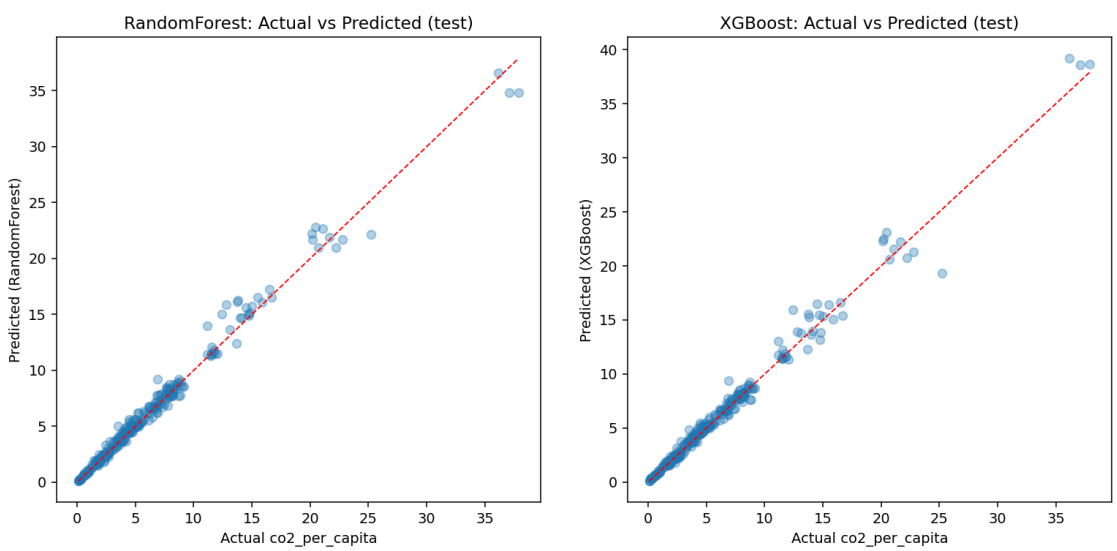


Figure 17: Random Forest (left) and XGBoost (right): Actual vs Predicted

Random Forest (RMSE = 0.682) and XGBoost (RMSE = 0.725) underperformed the linear baselines on this dataset. The CO_2 per capita–feature relationship is largely linear ($R^2 = 0.99$), and the non-linear interactions that tree models are designed to exploit are weak here. Additionally, COVID-19's structural break in the test period likely hurt tree-based models more than regularised linear models, as trees tend to extrapolate poorly. Tree-based models underperform compared to linear models, with higher variance and larger deviations at extreme values. This suggests that the underlying relationships are largely linear and do not benefit from nonlinear partitioning.

7.7 Deep Learning and Pretrained Models

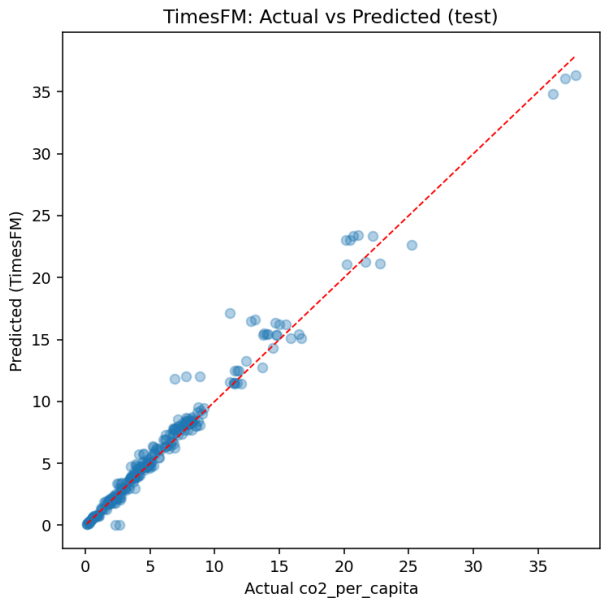


Figure 18: TimesFM: Actual vs Predicted CO_2 per capita

Deep learning models (MLP and LSTM) perform worse than linear baselines, likely due to limited data and weak temporal structure. MLP (RMSE = 0.782): Introduces unnecessary non-linearity and is prone to overfitting on tabular panel data without sufficient regularisation. LSTM (RMSE = 0.888): Sequence modelling is theoretically appropriate for time-series, but the short per-country sequences ($\leq$ 26 years) and the COVID structural break in the test period limit LSTM's advantage. The model is trained on train+val combined sequences, making an honest validation metric harder to compute.

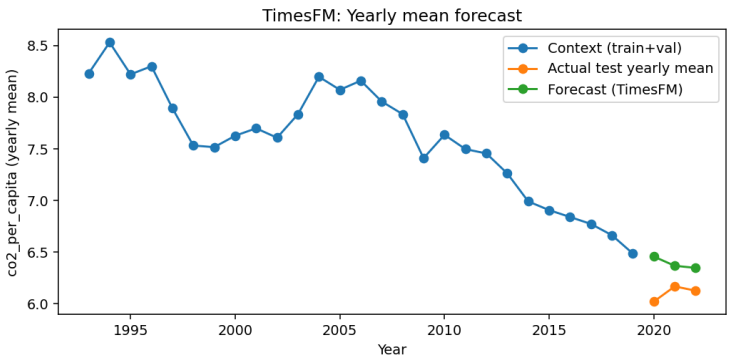


Figure 19: TimesFM yearly forecast vs actual values

Both zero-shot models perform substantially worse than trained linear models:

- PatchTST (RMSE $\approx$ 0.901): Fine-tuned on electricity data; mismatch with per-capita CO_2 (scale and multi-year horizon) leads to poor zero-shot generalization.
- TimesFM (RMSE $\approx$ 0.947): Google's 200M-parameter model shows similarly weak results; without domain-specific fine-tuning, scale and distribution differences dominate.

These results show that large pretrained time-series models do not automatically outperform lightweight domain-specific models, especially when the target domain differs from the pretraining data.

7.8 Key Findings

- Ridge regression is the best-performing model (RMSE = 0.5989)
- All linear models perform nearly identically, indicating strong linear structure
- Ensemble methods provide minimal improvement over individual models
- Tree-based and deep learning models underperform due to limited data complexity
- Pretrained models perform reasonably well but do not surpass classical approaches
- Distribution shift (COVID period) negatively impacts generalization of complex models

7.9 Conclusion

The results demonstrate that, despite the availability of advanced machine learning and deep learning techniques, regularized linear regression remains the most effective approach for forecasting CO_2 emissions in this dataset. This suggests that the dominant relationships are stable, linear, and well captured by simple models, while more complex methods introduce unnecessary variance without improving predictive performance.

8 R4: Economic Development, Energy Intensity, and CO_2 Emissions

8.1 Objective

R4: Do countries become more energy-efficient as they develop, and how does this affect CO_2 emissions over time?.
This analysis extends the project beyond prediction and clustering by examining a deeper development question. Instead of only asking whether variables are associated with emissions, this section studies whether economic development is linked to lower energy intensity and whether this, in turn, changes CO2 emissions per capita.
The analysis addresses three specific ideas:

1. As GDP increases, do countries tend to use less energy per unit of output?
2. Is higher energy intensity associated with higher CO2 emissions per capita?
3. Does the effect of GDP on CO2 emissions depend on energy intensity?

These questions matter because they help distinguish whether emissions are driven more by economic scale or by the way energy is used.

8.2 Exploratory Analysis

Figure 20 shows a negative relationship between GDP and energy intensity, suggesting that more developed economies tend to use less energy per unit of output. This supports the hypothesis that economic development is associated with improved energy efficiency.

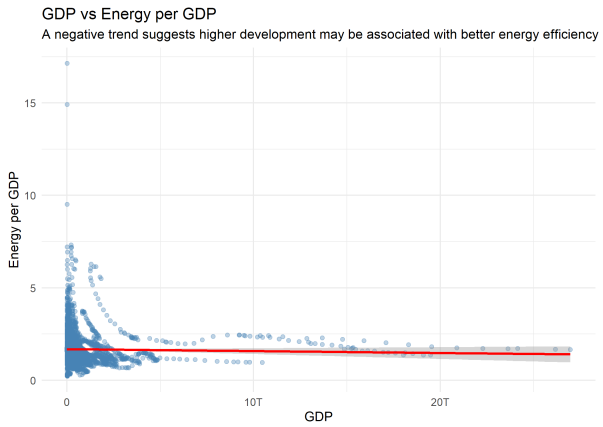


Figure 20: GDP vs Energy per GDP

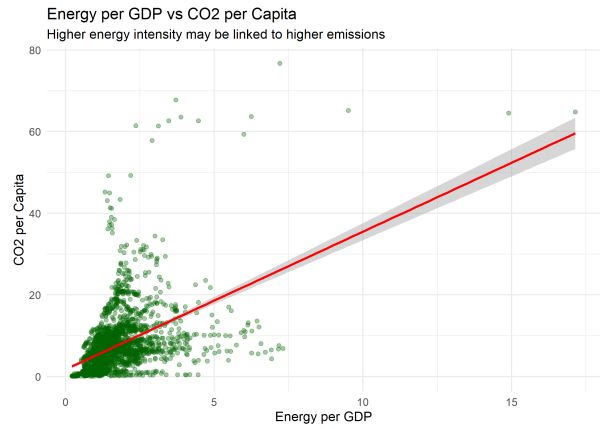


Figure 21: Energy per GDP vs CO_2 per capita

Figure 21 shows a strong positive relationship between energy intensity and CO_2 emissions per capita, indicating that countries using more energy per unit of GDP tend to have higher emissions.

To examine heterogeneity, Figure 22 groups countries by energy intensity. The differing slopes suggest that while GDP is positively associated with emissions, the magnitude varies across efficiency levels.

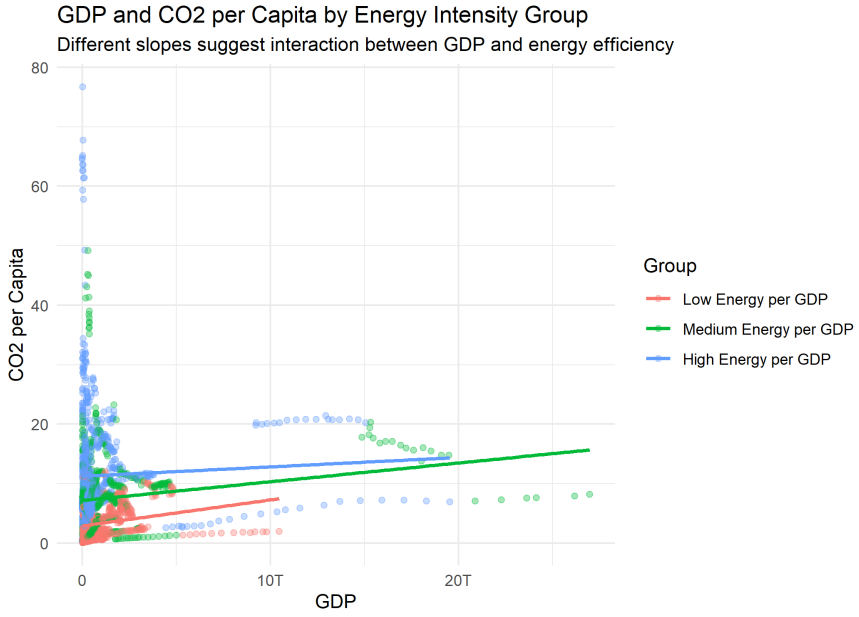


Figure 22: GDP and CO_2 per capita by energy intensity group

8.3 Trends Over Time

Aggregate trends provide additional insight into long-run dynamics. As shown in Figure 23, global GDP increases steadily over time, while energy intensity declines. At the same time, average CO_2 emissions per capita show a gradual downward trend after the mid-2000s.

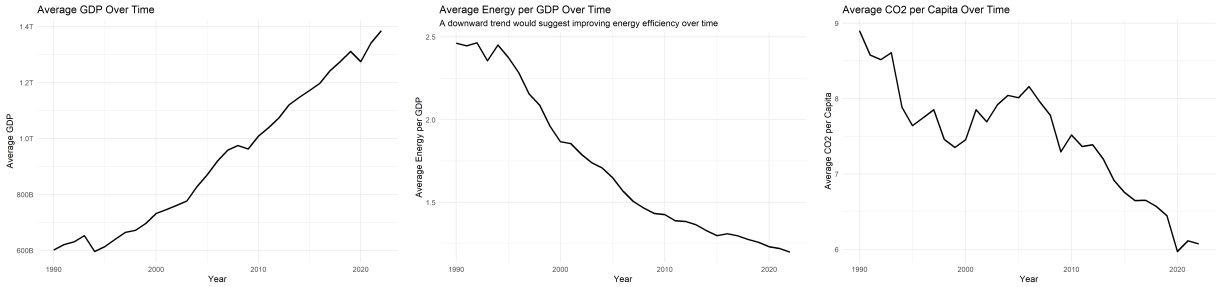


Figure 23: Trends over time: GDP (left), energy intensity (center), CO_2 per capita (right)

These patterns suggest that improvements in energy efficiency may partially offset the emissions associated with economic growth.

8.4 Model Overview

We estimate five models to examine the relationship between economic development, energy intensity, and CO_2 emissions:

- **Model 1:** Baseline OLS with GDP and energy intensity
- **Model 2:** Adds controls (population and energy use per capita)
- **Model 3:** Includes interaction between GDP and energy intensity
- **Model 4:** Fixed effects panel model (within-country variation)
- **Model 5:** Environmental Kuznets Curve (nonlinear GDP effect)

8.5 Regression Results

The baseline model confirms that both GDP and energy intensity are positively associated with CO_2 emissions per capita. However, explanatory power is limited when only these variables are included (Adjusted $R^2 \approx 0.225$).
After adding controls, model performance improves substantially (Adjusted $R^2 \approx 0.793$), indicating that energy use per capita is a key driver of emissions. The coefficient on energy per capita is large and highly significant, suggesting that emissions are primarily driven by how much energy is consumed rather than economic scale alone.
Energy intensity remains statistically significant across all specifications, reinforcing its role as an important determinant of emissions.

8.6 Interaction and Nonlinearity

The interaction model does not find a statistically significant interaction between GDP and energy intensity. This suggests that the effect of GDP on emissions does not strongly depend on efficiency levels in this specification.
However, the Environmental Kuznets Curve (EKC) model provides evidence of nonlinearity. The positive coefficient on GDP and negative coefficient on GDP^2 indicate an inverted-U relationship, where emissions rise during early development stages and decline at higher income levels.

8.7 Fixed Effects Analysis

The fixed effects panel model confirms that the positive relationship between GDP, energy intensity, and emissions also holds within countries over time. This strengthens the results by controlling for unobserved country-specific characteristics such as geography and long-term industrial structure.

8.8 Model Diagnostics

Diagnostic plots (Figure 24) reveal some deviations from ideal assumptions. Residuals exhibit heteroskedasticity and slight non-normality, particularly in the tails, and a small number of influential observations are present.

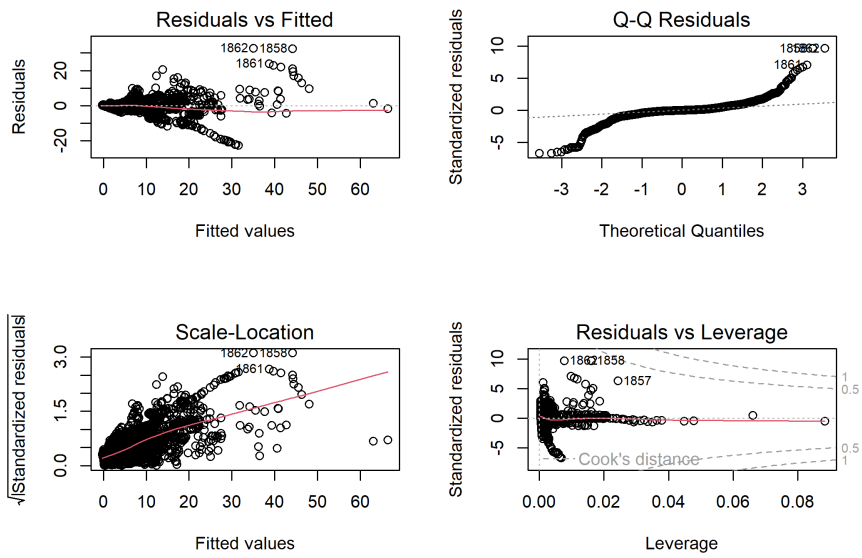


Figure 24: Regression diagnostics

Despite these issues, variance inflation factors remain low, indicating that multicollinearity is not a major concern.

8.9 Key Findings

- Economic development is associated with improved energy efficiency
- Higher energy intensity leads to significantly higher CO_2 emissions
- Energy use per capita is the strongest predictor of emissions
- No strong evidence that energy intensity moderates the GDP-emissions relationship
- Evidence supports an Environmental Kuznets Curve pattern
- Results are consistent both across countries and within countries over time

8.10 Conclusion

The analysis shows that while economic growth is associated with higher emissions, the primary driver is energy consumption rather than GDP itself. Improvements in energy efficiency play a meaningful role in reducing emissions, but are not sufficient on their own.
These findings suggest that policies targeting energy use and efficiency are likely to be more effective than those focused solely on limiting economic growth.

9 Model Evaluation Summary

Across all experiments, model performance was evaluated using Root Mean Squared Error (RMSE) on the test set. Among traditional machine learning models, regularized linear models performed best, with Ridge regression achieving the lowest RMSE (0.5989), followed closely by ElasticNet and standard linear regression.
Ensemble methods provided only marginal improvements over single models, suggesting that the underlying relationship between predictors and CO_2 emissions is largely linear. More complex models such as Random Forest and XGBoost did not outperform linear models, indicating limited nonlinear structure in the data.
Deep learning models (MLP and LSTM) performed worse than simpler approaches, likely due to the relatively small dataset and lack of strong temporal dependencies. Similarly, transformer-based models (PatchTST and TimesFM) showed the weakest performance, suggesting that these architectures are not well-suited for this problem setting.
Overall, the results indicate that simpler, interpretable models are sufficient and often preferable for predicting CO_2 emissions per capita in this dataset.

10 Results and Discussion

The analysis reveals several consistent patterns across exploratory analysis and modeling. First, GDP is positively associated with CO_2 emissions per capita, confirming that economic growth is generally linked to higher emissions. However, this relationship is moderated by energy efficiency.
Energy intensity (energy use per unit of GDP) is a key determinant of emissions. Countries with higher energy intensity tend to exhibit significantly higher CO_2 emissions, while those with lower energy intensity demonstrate more efficient and less emission-intensive growth.
Energy use per capita emerges as the strongest predictor of emissions, suggesting that absolute energy consumption plays a more critical role than economic output alone. This highlights the importance of energy systems and consumption patterns in driving environmental outcomes.
The Environmental Kuznets Curve (EKC) model provides evidence of a nonlinear relationship between GDP and emissions. Specifically, emissions increase during early stages of development but begin to decline at higher income levels. This suggests that advanced economies may benefit from cleaner technologies and stricter environmental policies.
Despite this, the interaction between GDP and energy intensity was not statistically significant, indicating that the effect of economic growth on emissions does not strongly depend on efficiency levels in this specification.

11 Conclusion

This study demonstrates that while economic growth is associated with increased CO_2 emissions, the primary driver is energy consumption rather than GDP itself. Improvements in energy efficiency contribute to reducing emissions but are not sufficient on their own.

The findings suggest that policies targeting energy consumption and promoting cleaner energy sources are likely to be more effective than those focused solely on limiting economic growth. Encouraging technological innovation, improving energy efficiency, and transitioning to low-carbon energy systems are critical for sustainable development.

12 Limitations

Several limitations should be acknowledged. First, the dataset size is relatively small, which may limit the performance of more complex models such as deep learning architectures. Second, the analysis relies on observational data, which restricts causal interpretation of the relationships identified.

Additionally, potential measurement errors in variables such as energy use and emissions may affect the results. The models also do not fully capture all relevant factors influencing emissions, such as policy differences, technological advancements, and sectoral composition of economies.

Finally, some model assumptions, including homoskedasticity and normality of residuals, are not perfectly satisfied, which may impact inference.

13 Appendix

13.1 Model List

- Model 1: Baseline OLS (GDP, Energy Intensity)
- Model 2: OLS with controls (Population, Energy Use per Capita)
- Model 3: Interaction model (GDP $\times$ Energy Intensity)
- Model 4: Fixed Effects panel model
- Model 5: Environmental Kuznets Curve (GDP and GDP^2)

13.2 Evaluation Metric

Model performance is evaluated using Root Mean Squared Error (RMSE):

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

where y_i represents actual values and $\hat{y}_i$ represents predicted values.

13.3 Software and Libraries

All analyses were conducted using R and Python.

R Libraries:

- tidyverse
- ggplot2
- dplyr
- glmnet (Ridge, Lasso, ElasticNet)
- caret (model training and evaluation)
- corrplot (correlation analysis)
- cluster, factoextra (clustering methods)
- dbscan (density-based clustering)
- mclust (Gaussian Mixture Models)
- plm (panel data models for R4)
- car (diagnostics and VIF)

Python Libraries:

- numpy, pandas
- scikit-learn (ML models)

- xgboost
- tensorflow / keras (LSTM)
- pytorch (deep learning models)
- transformers / huggingface (PatchTST, TimesFM)